GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Atrial Cardiac Muscle Cell Action Potent	−0.36823663	11	2.351e−05	6.333e−02	CACNB2:58 GJC1:243 SCN5A:536 KCNN2:896 GJA1:1171 KCNE5:1480
Cardiac Muscle Cell Action Potential Inv	−0.23434356	28	1.782e−05	6.333e−02	KCNH2:189 GJC1:243 CACNA1G:461 SCN5A:536 KCNE2:632 GJA1:1171
NADH Dehydrogenase Complex Assembly (GO:	0.16169979	48	1.074e−04	7.236e−02	NDUFA11:113 NDUFB8:126 NDUFAF1:134 NDUFS5:238 OXA1L:251 NDUFS3:301
Mitochondrial Respiratory Chain Complex	0.16169979	48	1.074e−04	7.236e−02	NDUFA11:113 NDUFB8:126 NDUFAF1:134 NDUFS5:238 OXA1L:251 NDUFS3:301
Positive Regulation Of Tyrosine Phosphor	0.16341999	49	7.661e−05	7.236e−02	IL6ST:71 GH2:179 PARP9:221 IL4:226 CSF1R:243 CSH2:255
Potassium Ion Transmembrane Transport (G	−0.10283296	132	4.686e−05	7.236e−02	KCNA4:68 KCNF1:115 KCNV2:156 KCNH2:189 KCNK18:196 SLC12A4:199
Regulation Of Cardiac Muscle Cell Membra	−0.25179390	20	9.720e−05	7.236e−02	KCNH2:189 SCN5A:536 KCNE2:632 NOS1AP:886 KCNE5:1480 SNTA1:1551
Regulation Of Heart Rate By Cardiac Cond	−0.18757818	38	6.353e−05	7.236e−02	CACNB2:58 KCNH2:189 CACNA1G:461 SCN5A:536 KCNE2:632 KCND3:1022
Regulation Of Ventricular Cardiac Muscle	−0.25754099	17	2.371e−04	1.420e−01	KCNH2:189 SCN5A:536 KCNE2:632 NOS1AP:886 KCNE5:1480 SNTA1:1551
Oxidative Phosphorylation (GO:0006119)	0.14195122	55	2.741e−04	1.477e−01	NDUFA11:113 NDUFB8:126 NDUFS5:238 NDUFS3:301 ATP5PD:425 NDUFA12:723
Potassium Ion Transport (GO:0006813)	−0.09573849	119	3.178e−04	1.556e−01	KCNA4:68 KCNF1:115 KCNV2:156 KCNH2:189 KCNK18:196 SLC12A4:199
Cardiac Conduction (GO:0061337)	−0.15636386	43	3.917e−04	1.623e−01	CACNB2:58 KCNH2:189 CACNA1G:461 SCN5A:536 KCNE2:632 KCND3:1022
Proton Motive Force−Driven Mitochondrial	0.15037531	47	3.648e−04	1.623e−01	NDUFA11:113 NDUFB8:126 NDUFS5:238 NDUFS3:301 ATP5PD:425 NDUFA12:723
Potassium Ion Export Across Plasma Membr	−0.29074885	12	4.882e−04	1.754e−01	KCNH2:189 KCNK18:196 KCNE2:632 KCND3:1022 KCNE5:1480 KCNE1:1691
Ventricular Cardiac Muscle Cell Action P	−0.25304333	16	4.585e−04	1.754e−01	KCNH2:189 SCN5A:536 KCNE2:632 KCNE5:1480 SNTA1:1551 NEDD4L:1603
Atrial Cardiac Muscle Cell To AV Node Ce	−0.40231686	6	6.430e−04	2.165e−01	GJC1:243 SCN5A:536 GJA1:1171 KCNE5:1480 SCN3B:2699 KCNQ1:3457
Mitochondrial Electron Transport, NADH T	0.17567919	31	7.143e−04	2.264e−01	NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFB5:783 NDUFB10:1188
Cellular Respiration (GO:0045333)	0.11556861	71	7.694e−04	2.303e−01	NDUFA11:113 NDUFB8:126 NDUFS5:238 OXA1L:251 NDUFS3:301 BLOC1S1:507
Aerobic Respiration (GO:0009060)	0.13175380	54	8.184e−04	2.321e−01	NDUFA11:113 NDUFB8:126 NDUFS5:238 OXA1L:251 NDUFS3:301 BLOC1S1:507
Chemical Synaptic Transmission (GO:00072	−0.06000401	253	1.067e−03	2.640e−01	RIMBP2:9 CHRNB2:19 HTR3B:41 CHRNA3:48 CACNB2:58 GRIN3B:62
Monoatomic Cation Transmembrane Transpor	−0.05803821	273	1.019e−03	2.640e−01	ATP6V0A2:6 ATP2A1:15 TRPM5:16 KCNA4:68 KCNF1:115 KCNV2:156
Regulation Of Rho Protein Signal Transdu	−0.11476776	68	1.078e−03	2.640e−01	GPR35:99 EPS8L3:136 GPR20:247 EPS8L1:279 AKAP13:390 RIPOR2:440
Regulation Of Smooth Muscle Contraction	−0.20922291	20	1.201e−03	2.814e−01	CHRNA3:48 TACR2:121 PROK2:493 SETD3:942 CHRM3:1081 CTTN:1395
Regulation Of Tyrosine Phosphorylation O	0.12365420	57	1.253e−03	2.814e−01	IL6ST:71 GH2:179 PARP9:221 IL4:226 CSF1R:243 CSH2:255
Neuropeptide Signaling Pathway (GO:00072	−0.11716023	63	1.313e−03	2.830e−01	SSTR4:13 PENK:270 OPRK1:367 NTSR2:519 RXFP3:737 OPRL1:848
Cardiac Muscle Cell Action Potential (GO	−0.17080375	29	1.460e−03	3.026e−01	KCNH2:189 CACNA1G:461 SCN5A:536 KCNE2:632 KCND3:1022 HCN2:1177
Regulatory ncRNA Processing (GO:0070918)	−0.16194314	32	1.529e−03	3.050e−01	MOV10L1:104 TDRKH:129 PIWIL1:195 ANKRD34C:264 MAEL:680 ANKRD34B:1025
Proton Motive Force−Driven ATP Synthesis	0.12689437	51	1.732e−03	3.334e−01	NDUFA11:113 NDUFB8:126 NDUFS5:238 NDUFS3:301 ATP5PD:425 NDUFA12:723
Intermediate Filament Organization (GO:0	−0.12718007	50	1.878e−03	3.373e−01	KRT15:222 KRT19:327 KRT78:395 KRT80:602 KRT36:730 KRT35:825
piRNA Processing (GO:0034587)	−0.21194938	18	1.854e−03	3.373e−01	MOV10L1:104 TDRKH:129 PIWIL1:195 ANKRD34C:264 MAEL:680 ANKRD34B:1025
Ventricular Cardiac Muscle Cell Membrane	−0.33207627	7	2.346e−03	4.077e−01	KCNH2:189 KCNE2:632 KCNJ3:1527 KCNE1:1691 KCNJ5:2499 KCNQ1:3457
Anterograde Trans−Synaptic Signaling (GO	−0.06498271	180	2.717e−03	4.351e−01	CHRNB2:19 HTR3B:41 CHRNA3:48 CACNB2:58 HTR1D:75 OMP:77
Mitochondrial Respiratory Chain Complex	0.09883365	77	2.746e−03	4.351e−01	NDUFA11:113 NDUFB8:126 NDUFAF1:134 NDUFS5:238 OXA1L:251 NDUFS3:301
tRNA Methylation (GO:0030488)	0.14461673	36	2.688e−03	4.351e−01	TRMT13:300 TRMT10A:313 BCDIN3D:386 TRMO:499 THUMPD2:502 TRMT6:527
Regulation Of Dendrite Development (GO:0	−0.13940076	38	2.958e−03	4.553e−01	CHRNB2:19 CHRNA3:48 MFSD2A:50 COBL:378 RAB17:383 KNDC1:1187
G Protein−Coupled Receptor Signaling Pat	−0.12493053	47	3.064e−03	4.586e−01	HTR1D:75 OR10J5:124 AGTR2:469 PTGIR:515 HTR6:594 MTNR1A:808
Aerobic Electron Transport Chain (GO:001	0.11559099	54	3.325e−03	4.594e−01	NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFB5:783 COX6B1:791
Cardiac Muscle Cell Contraction (GO:0086	−0.19992940	18	3.324e−03	4.594e−01	GSN:237 CACNA1G:461 SCN5A:536 KCNE2:632 PIK3CA:1557 KCNE1:1691
Mitochondrial ATP Synthesis Coupled Elec	0.11475769	55	3.266e−03	4.594e−01	NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFA12:723 NDUFB5:783
Nitric Oxide Mediated Signal Transductio	−0.20455062	17	3.506e−03	4.702e−01	AGTR2:469 NEUROD1:488 CD36:1259 RILPL1:1688 NOS1:1957 DDAH2:2049

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
BENPORATH_ES_WITH_H3K27ME3	−0.05889188	976	6.811e−10	4.419e−06	CEMIP:20 SUSD4:31 PLEC:35 LY6H:53 ADRB1:57 ZBTB16:60
MARTENS_TRETINOIN_RESPONSE_UP	−0.06874780	675	1.434e−09	4.651e−06	INHBB:47 LY6H:53 FGF8:61 GRIN3B:62 CEP170B:73 GPR35:99
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	−0.15109777	127	4.245e−09	9.181e−06	HELZ2:59 MTG2:171 ABHD16B:212 FAM217B:224 SLCO4A1:245 SYCP2:306
HAMAI_APOPTOSIS_VIA_TRAIL_UP	0.06987262	591	7.861e−09	1.275e−05	BAZ2B:34 PTPN13:53 MIS18BP1:68 ANKHD1:82 QSER1:86 CENPC:93
DODD_NASOPHARYNGEAL_CARCINOMA_DN	0.04798771	1204	3.013e−08	3.910e−05	TET1:24 CD80:39 QSER1:86 CDC7:89 URI1:91 PPRC1:101
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	−0.07616601	419	9.955e−08	9.274e−05	CHRNB2:19 CHRNA3:48 ZBTB16:60 FGF8:61 GRIN3B:62 HTR1D:75
DACOSTA_UV_RESPONSE_VIA_ERCC3_DN	0.05552342	809	1.001e−07	9.274e−05	ZFYVE9:4 ATP2B4:7 UBR2:8 AKAP9:15 BAZ2B:34 FAM169A:36
BENPORATH_SUZ12_TARGETS	−0.04849790	912	8.536e−07	5.538e−04	CEMIP:20 SUSD4:31 PLEC:35 ITPKB:42 CHRNA3:48 ADRB1:57
MIKKELSEN_NPC_HCP_WITH_H3K27ME3	−0.07943135	328	8.204e−07	5.538e−04	LY6H:53 BNC1:130 S1PR5:154 PENK:270 DLGAP2:277 C1QL2:284
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT	−0.08369977	296	7.787e−07	5.538e−04	SSTR4:13 ADRB1:57 GPR143:63 HTR1D:75 GPR35:99 GPR132:100
WP_GPCRS_CLASS_A_RHODOPSINLIKE	−0.09685368	211	1.296e−06	7.646e−04	OR6B1:8 SSTR4:13 ADRB1:57 HTR1D:75 GPR35:99 P2RY4:135
FISCHER_DREAM_TARGETS	0.04707441	865	3.148e−06	1.702e−03	ORC1:30 DNMT3B:54 MIS18BP1:68 CDC7:89 CENPC:93 RTKN2:95
MEISSNER_BRAIN_HCP_WITH_H3K4ME3_AND_H3K2	−0.04330605	1019	3.666e−06	1.830e−03	CHRNB2:19 CEMIP:20 MINAR1:27 ITPKB:42 INHBB:47 MFSD2A:50
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	−0.05704919	550	5.313e−06	2.462e−03	CHRNB2:19 RASGRP2:23 CHRNA3:48 LY6H:53 GRIN3B:62 CPNE7:95
BENPORATH_EED_TARGETS	−0.04367894	917	8.792e−06	3.803e−03	CEMIP:20 SUSD4:31 PLEC:35 INHBB:47 ADRB1:57 ZBTB16:60
REACTOME_SIGNALING_BY_GPCR	−0.05154213	623	1.263e−05	5.122e−03	SSTR4:13 RASGRP2:23 ADRB1:57 GPR143:63 HTR1D:75 GPR35:99
JOHNSTONE_PARVB_TARGETS_3_DN	0.04564997	766	1.976e−05	7.490e−03	UBAP2:66 MIS18BP1:68 SENP7:70 QSER1:86 RTKN2:95 N4BP2:97
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	−0.07670211	261	2.078e−05	7.490e−03	SSTR4:13 CHRNB2:19 CHRNA3:48 ADRB1:57 GRIN3B:62 HTR1D:75
BENPORATH_PRC2_TARGETS	−0.05169699	576	2.468e−05	7.625e−03	CEMIP:20 SUSD4:31 PLEC:35 ADRB1:57 ZBTB16:60 BNC1:130
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	0.05130001	586	2.442e−05	7.625e−03	GOT1:60 IL6ST:71 ZNF644:105 APC:116 WDR76:136 HPS3:145
DACOSTA_UV_RESPONSE_VIA_ERCC3_COMMON_DN	0.05884521	441	2.447e−05	7.625e−03	ZFYVE9:4 UBR2:8 AKAP9:15 BAZ2B:34 FAM169A:36 VPS13A:48
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	−0.06709733	330	2.950e−05	8.531e−03	LY6H:53 FGF8:61 BNC1:130 PENK:270 DLGAP2:277 C1QL2:284
GEORGES_TARGETS_OF_MIR192_AND_MIR215	0.04353486	807	3.024e−05	8.531e−03	ORC1:30 DST:49 MIS18BP1:68 IL6ST:71 USP54:80 CDC7:89
REACTOME_SPERM_MOTILITY_AND_TAXES	0.42411889	8	3.263e−05	8.566e−03	CATSPERB:10 CATSPER2:19 CATSPERG:32 CATSPERD:124 KCNU1:127 CATSPER4:683
KEGG_ALLOGRAFT_REJECTION	0.25570555	22	3.301e−05	8.566e−03	CD86:18 CD80:39 IL4:226 CD28:411 IL12B:554 IFNG:558
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	0.16796433	47	6.811e−05	1.700e−02	NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFA12:723 NDUFB5:783
REACTOME_GPCR_LIGAND_BINDING	−0.05778646	398	8.077e−05	1.941e−02	SSTR4:13 ADRB1:57 GPR143:63 HTR1D:75 GPR35:99 GPR132:100
NIKOLSKY_BREAST_CANCER_21Q22_AMPLICON	−0.29774673	14	1.146e−04	2.656e−02	COL6A1:172 SPATC1L:226 COL6A2:278 POFUT2:335 COL18A1:405 DIP2A:417
MEISSNER_BRAIN_HCP_WITH_H3K27ME3	−0.06972794	257	1.230e−04	2.752e−02	CHRNA3:48 FGF8:61 GRIN3B:62 COL8A2:90 BNC1:130 ALX4:210
NIKOLSKY_BREAST_CANCER_1Q21_AMPLICON	−0.19210430	33	1.342e−04	2.796e−02	TDRKH:129 PMVK:482 PYGO2:644 RORC:707 PKLR:828 ADAM15:837
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	−0.28412760	15	1.390e−04	2.796e−02	SFSWAP:79 GALNT9:219 EP400:398 GOLGA3:756 NOC4L:1145 DDX51:1249
REACTOME_COMPLEX_I_BIOGENESIS	0.15880372	48	1.416e−04	2.796e−02	NDUFA11:113 NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFA12:723
PUJANA_BRCA1_PCC_NETWORK	0.03049273	1428	1.422e−04	2.796e−02	ORC1:30 IRAG2:35 TMEM131L:55 CDC7:89 CENPC:93 UBR5:104
NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON	−0.16103940	46	1.582e−04	3.018e−02	PIEZO1:32 ANKRD11:92 CPNE7:95 SPATA2L:206 ZC3H18:215 CBFA2T3:246
BIOCARTA_TH1TH2_PATHWAY	0.24835167	19	1.785e−04	3.309e−02	CD86:18 IL18R1:218 IL4:226 CD28:411 IL12B:554 IFNG:558
YOSHIMURA_MAPK8_TARGETS_UP	−0.03373108	1102	1.846e−04	3.328e−02	SSTR4:13 CHRNB2:19 ITPKB:42 H1−6:43 INHBB:47 ADRB1:57
REACTOME_FERTILIZATION	0.22882170	21	2.836e−04	4.842e−02	CATSPERB:10 CATSPER2:19 CATSPERG:32 CATSPERD:124 KCNU1:127 CD9:263
PUJANA_ATM_PCC_NETWORK	0.03144663	1205	2.810e−04	4.842e−02	UBR2:8 IRAG2:35 TMEM131L:55 CDC7:89 CENPC:93 APC:116
CHARAFE_BREAST_CANCER_LUMINAL_VS_MESENCH	−0.05271653	404	2.917e−04	4.853e−02	CAPN8:44 INHBB:47 PER2:72 FAM83H:109 MB:126 SPATA2L:206
BIOCARTA_LAIR_PATHWAY	0.26901306	15	3.094e−04	4.895e−02	VCAM1:125 KNG1:141 ICAM1:450 ITGA4:643 TNF:1046 ITGAL:1202

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Long QT Syndrome	−0.14307558	58	1.660e−04	8.142e−01	KCNA4:68 KCNH2:189 CUZD1:331 MINK1:368 RPGR:509 DNM2:526
Specific learning disability	0.12057948	86	1.132e−04	8.142e−01	NSD1:41 TINF2:289 DGCR6:295 SPG11:439 POMK:520 LTBP4:553
Acidosis, Lactic	0.05094047	150	3.168e−02	9.709e−01	NDUFA11:113 NDUFAF1:134 LRPPRC:162 TANGO2:204 G6PC:213 HSD17B6:296
Aneurysm, Dissecting	−0.15189697	18	2.570e−02	9.709e−01	CDKN2B:85 LOXL1:311 PLOD1:655 NOS3:2058 APOE:2117 MTHFR:2815
Angioedemas, Hereditary	0.15916457	16	2.753e−02	9.709e−01	KNG1:141 C4B:637 ACE:811 A2M:832 SERPINA5:1476 SERPINA4:1575
ATRIAL FIBRILLATION, FAMILIAL, 1 (disord	−0.14936336	20	2.078e−02	9.709e−01	SCN5A:536 KCNE2:632 KCNE1:1691 GATA5:2027 SCN2B:2258 NPPA:2588
Brain Injuries, Focal	0.08483396	59	2.432e−02	9.709e−01	TNFSF10:174 IL18R1:218 TLR1:327 CHRNA7:449 IL31RA:546 IL1R1:899
Congenital secretory diarrhea, sodium ty	−0.24229406	7	2.642e−02	9.709e−01	SLC9A3:1153 SLC9A2:1767 SPINT2:2175 HGF:2543 CAV1:3333 SLC9A1:8951
Dermatitis, Atopic	0.03606425	375	1.716e−02	9.709e−01	SLCO6A1:37 ALDH1A1:57 VCAM1:125 DNMT1:155 CXCR2:212 IL18R1:218
Diabetes Mellitus, Insulin−Dependent	0.02385339	736	2.956e−02	9.709e−01	CUX2:16 CD86:18 SLCO6A1:37 CD80:39 G6PC2:87 APC:116
Empyema, Pleural	0.31604773	6	7.340e−03	9.709e−01	SMC2:529 NFKBIE:719 TNFRSF6B:1126 SMC4:1221 NFKBIA:7017 NFKBIB:7017
EPILEPSY, PYRIDOXINE−DEPENDENT	0.19484906	11	2.525e−02	9.709e−01	RNF19A:631 ALDH7A1:737 PLPBP:1508 CRK:1612 GRAP2:7017 MAPK14:7017
FANCONI ANEMIA, COMPLEMENTATION GROUP A	0.04640449	214	1.972e−02	9.709e−01	AFP:14 RNF168:46 TET2:50 ALDH1A1:57 BRIP1:109 MEFV:118
HYPERTRICHOSIS, CONGENITAL GENERALIZED	0.17354073	14	2.457e−02	9.709e−01	IGF1:377 FGFR1:382 ABCC9:398 BRD2:819 BRCA1:1157 SOX3:7017
Ichthyosis, X−Linked	0.08977444	82	5.002e−03	9.709e−01	TNFSF10:174 ABCB1:177 IGF1:377 EGF:522 MDM4:564 KDR:709
Immunodeficiency syndrome, variable	0.36721620	5	4.458e−03	9.709e−01	DNMT3B:54 CDCA7:766 ATRX:1186 ZBTB24:1600 HELLS:7017 NA
Infant, Premature	0.04636799	216	1.926e−02	9.709e−01	SLCO6A1:37 DNMT3B:54 MBD1:79 LCT:83 ABCB1:177 TSLP:273
Injuries, Acute Brain	0.08483396	59	2.432e−02	9.709e−01	TNFSF10:174 IL18R1:218 TLR1:327 CHRNA7:449 IL31RA:546 IL1R1:899
Leprosy, Lepromatous	0.12408984	33	1.366e−02	9.709e−01	CD80:39 CD28:411 ICAM1:450 FOXP3:469 IL12B:554 IL1R1:899
Leukemia, Myelocytic, Acute	0.02195948	1365	7.917e−03	9.709e−01	CD86:18 TET1:24 SLCO6A1:37 SLC22A16:38 CD80:39 NSD1:41
Lupus Erythematosus, Subacute Cutaneous	0.17516710	17	1.242e−02	9.709e−01	MBD1:79 DNMT1:155 CD28:411 MBD4:517 RBM45:1027 TNF:1046
Lupus Erythematosus, Systemic	0.03485716	844	6.998e−04	9.709e−01	CD86:18 SLCO6A1:37 CD80:39 DNMT3B:54 PLA2R1:64 RTKN2:95
Lymphoma, Non−Hodgkin	0.03763787	443	7.018e−03	9.709e−01	SLCO6A1:37 TET2:50 VCAM1:125 HDAC9:168 TNFSF10:174 ABCB1:177
Lymphoma, T−Cell, Cutaneous	0.05008112	210	1.266e−02	9.709e−01	CD80:39 STAT2:114 DNMT1:155 HDAC9:168 TNFSF10:174 ABCB1:177
Multiple Sclerosis, Primary Progressive	0.21676745	10	1.762e−02	9.709e−01	CSF1R:243 IL7R:772 CASP8:962 RBM45:1027 TNF:1046 LAMC2:7017
Neurogenic muscle atrophy, especially in	0.05251235	196	1.151e−02	9.709e−01	VPS13A:48 NDUFA11:113 COL2A1:129 NDUFAF1:134 COL6A3:235 NEK1:278
Pain, Postoperative	0.15440444	21	1.433e−02	9.709e−01	TRPV1:165 ABCB1:177 SCN11A:388 LIF:813 TNF:1046 CRHR1:1641
Psychoses, Drug	−0.19192014	20	2.971e−03	9.709e−01	CALR:180 DTNBP1:187 RGS9:292 AKT1:352 HTR6:594 CAMK2B:1605
Psychoses, Substance−Induced	−0.18622319	16	9.918e−03	9.709e−01	RGS9:292 AKT1:352 HTR6:594 CAMK2B:1605 PHB:1996 ESR1:2263
Schizoaffective disorder, bipolar type	0.22456067	9	1.966e−02	9.709e−01	CHRNA7:449 DISC1:1133 CD40:1397 GABRA4:1549 NFKB1:7017 GABRB1:7017
Skin Diseases, Bullous	0.26331423	6	2.550e−02	9.709e−01	DST:49 COL7A1:96 RASA2:1497 RASA1:7017 IQGAP1:7017 DSG3:7017
Toxoplasmosis, Cerebral	0.32342781	6	6.077e−03	9.709e−01	IL6ST:71 LRPPRC:162 RBM45:1027 GFAP:1639 IL6:7017 CD40LG:7017
21−hydroxylase deficiency	0.14346283	19	3.042e−02	9.709e−01	HSD17B6:296 C4A:542 C4B:637 BRD2:819 GLO1:920 HLA−DOA:1282
22q11 partial monosomy syndrome	0.19576300	11	2.457e−02	9.709e−01	PI4KA:708 JMJD1C:986 TBX1:1039 HIRA:1567 COMT:7017 RREB1:7017
Abnormal mental state	0.24304083	8	1.729e−02	9.709e−01	SPG11:439 FMR1:887 CHL1:1659 GBA2:1793 CA2:7017 DYRK1A:7017
Abnormal mitochondria in muscle tissue	0.12402894	25	3.187e−02	9.709e−01	NDUFA11:113 NDUFAF1:134 NDUFS3:301 NDUFB10:1188 TMEM126B:1608 SDHA:7017
Abnormality of body height	−0.15248079	17	2.953e−02	9.709e−01	FGF8:61 WDR11:369 PROK2:493 GNRH1:533 TAC3:708 PROKR2:945
Abnormality of the outer ear	−0.09794543	45	2.309e−02	9.709e−01	PEX14:198 KIF1A:591 FGD1:722 DACT1:1107 CCBE1:1109 GJA1:1171
Absence of pubertal development	−0.15350986	21	1.490e−02	9.709e−01	FGF8:61 WDR11:369 PROK2:493 GNRH1:533 TAC3:708 PROKR2:945
Absent ear	0.24130698	7	2.704e−02	9.709e−01	ORC1:30 CHUK:303 CDC6:492 CDC45:7017 GMNN:7017 ORC6:7017

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0002429 abnormal blood cell	0.03451397	1183	2.886e−04	1.374e−01	CD86:18 CD80:39 VPS13A:48 PTPN13:53 DNMT3B:54 MAN2B1:59
MP0001800 abnormal humoral immune	0.05335221	326	1.226e−03	1.459e−01	CD86:18 CD80:39 IL6ST:71 TNFSF10:174 CSF2RB:187 CXCR2:212
MP0000716 abnormal immune system	0.03385226	965	9.837e−04	1.459e−01	CD86:18 CD80:39 PTPN13:53 DNMT3B:54 MAN2B1:59 IL6ST:71
MP0002723 abnormal immune serum	0.04203653	620	6.558e−04	1.459e−01	CD86:18 CD80:39 PTPN13:53 IL6ST:71 G6PC2:87 TRPV1:165
MP0005000 abnormal immune tolerance	0.05910516	239	1.981e−03	1.886e−01	CD86:18 CD80:39 IL6ST:71 TNFSF10:174 CXCR2:212 IL17RC:215
MP0003077 abnormal cell cycle	0.07116245	148	3.110e−03	1.967e−01	CUX2:16 CDC7:89 APC:116 FANCM:189 REL:321 PRDX1:339
MP0002080 prenatal lethality	0.02667289	1308	3.719e−03	1.967e−01	UBR2:8 CUX2:16 LRP4:17 NSD1:41 CHD8:42 DNMT3B:54
MP0004742 abnormal vestibular system	−0.13782173	39	2.975e−03	1.967e−01	PCDH15:3 AP3D1:362 SLC30A4:658 OTOP1:850 MBP:985 GRID2:1051
MP0008469 abnormal protein level	0.04448005	384	3.638e−03	1.967e−01	CD86:18 CXCR2:212 IL17RC:215 IL4:226 CSF1R:243 IFNAR1:283
MP0002452 abnormal antigen presenting	0.03630140	562	4.820e−03	2.295e−01	CD86:18 CD80:39 IL6ST:71 STAT2:114 MEFV:118 TNFSF10:174
MP0002398 abnormal bone marrow	0.03082621	788	5.673e−03	2.455e−01	CD86:18 CD80:39 DST:49 DNMT3B:54 IL6ST:71 G6PC2:87
MP0009115 abnormal fat cell	−0.08232819	85	9.108e−03	3.335e−01	ADRB1:57 LPIN1:165 PLSCR3:302 PRLH:326 SLC2A4:345 AGTR2:469
MP0002420 abnormal adaptive immunity	0.02835766	852	8.610e−03	3.335e−01	CD86:18 CD80:39 PTPN13:53 IL6ST:71 G6PC2:87 STAT2:114
MP0004085 abnormal heartbeat	−0.05956181	156	1.113e−02	3.783e−01	ADRB1:57 CACNB2:58 ENPP2:76 KCNH2:189 BDKRB1:324 CACNA1G:461
MP0000689 abnormal spleen morphology	0.03312021	532	1.207e−02	3.830e−01	CD86:18 CD80:39 MAN2B1:59 IL6ST:71 DNMT1:155 CXCR2:212
MP0001819 abnormal immune cell	0.02637456	855	1.441e−02	4.287e−01	CD86:18 CD80:39 PTPN13:53 IL6ST:71 G6PC2:87 STAT2:114
MP0000313 abnormal cell death	0.03450651	449	1.538e−02	4.307e−01	DNMT3B:54 UBR5:104 APC:116 DAXX:132 DNMT1:155 LAMA1:188
MP0001666 abnormal nutrient absorption	−0.10718577	42	1.655e−02	4.377e−01	TREH:203 SLC2A4:345 ABCB4:667 TRPV6:701 EPHB6:702 HSD17B4:846
MP0002069 abnormal eating/drinking beha	−0.03564006	397	1.798e−02	4.505e−01	PLEC:35 PER2:72 GHRL:161 ITGB4:193 HAX1:253 CCR5:273
MP0003137 abnormal impulse conducting	−0.08431368	64	2.022e−02	4.708e−01	EMD:105 GJC1:243 CACNA1G:461 SCN5A:536 FXN:966 GJA1:1171
MP0003635 abnormal synaptic transmissio	−0.03390430	421	2.077e−02	4.708e−01	CHRNB2:19 CHRNA3:48 ADORA1:163 BACE1:181 DVL1:244 PPP3R1:296
MP0002837 dystrophic cardiac calcinosis	−0.21773923	9	2.378e−02	4.921e−01	SLC2A4:345 DNM1L:678 MYBPC3:2077 DES:2483 MMP9:3155 SCG5:3598
MP0002722 abnormal immune system	0.02840070	602	2.304e−02	4.921e−01	ATP2B4:7 CD80:39 DNMT3B:54 IL6ST:71 APC:116 DNMT1:155
MP0001486 abnormal startle reflex	−0.06607896	95	2.705e−02	5.364e−01	PCDH15:3 OTOF:69 FIGN:202 DVL1:244 TMOD2:394 SCARB2:451
MP0005452 abnormal adipose tissue	−0.03574192	315	3.308e−02	6.298e−01	ADRB1:57 ADAMTS1:112 GPR68:142 GHRL:161 LPIN1:165 PNN:168
MP0005197 abnormal uvea morphology	−0.06630361	85	3.570e−02	6.535e−01	CHRNB2:19 CHRNA3:48 GPR143:63 PNN:168 DTNBP1:187 ATF4:374
MP0005671 abnormal response to	0.09492139	39	4.078e−02	6.694e−01	CD28:411 IFNG:558 TNFSF8:999 TNF:1046 CD8A:1185 CD40:1397
MP0001970 abnormal pain threshold	−0.04906579	150	4.018e−02	6.694e−01	SSTR4:13 CHRNB2:19 ADORA1:163 BACE1:181 PENK:270 BDKRB1:324
MP0000678 abnormal parathyroid gland	0.13089839	21	3.811e−02	6.694e−01	TBX1:1039 CASR:1086 HOXA3:1450 GCM2:1585 CYP27B1:1761 RARA:7017
MP0003448 altered tumor morphology	0.06397472	84	4.392e−02	6.969e−01	APC:116 TNFSF10:174 CXCR2:212 IFNAR1:283 PRDX1:339 IGF1:377
MP0002971 abnormal brown adipose	−0.06616886	76	4.729e−02	7.261e−01	ADRB1:57 GPR68:142 LPIN1:165 PNN:168 PRLH:326 LIPE:542
MP0000026 abnormal inner ear	−0.04124308	196	4.967e−02	7.389e−01	PCDH15:3 FGF8:61 OTOF:69 ATP8B1:120 FIGN:202 DVL1:244
MP0009278 abnormal bone marrow	−0.13665141	17	5.136e−02	7.408e−01	LIG1:234 EP400:398 SOCS3:732 ABL1:1435 RAG2:2938 HIP1:3054
MP0003656 abnormal erythrocyte physiolo	0.11337434	22	6.604e−02	7.668e−01	NEK1:278 MAN2A1:481 KLF1:895 NFE2L2:1150 SGMS2:7017 NCKAP1L:7017
MP0005253 abnormal eye physiology	−0.05667454	90	6.482e−02	7.668e−01	CACNB2:58 RGS9BP:179 GJC1:243 SEMA4A:283 TUB:435 NEUROD1:488
MP0000858 altered metastatic potential	0.07290594	58	5.579e−02	7.668e−01	CXCR2:212 CASP1:743 MTBP:1103 CDKN2A:1174 ITGAL:1202 MAP3K1:1506
MP0003942 abnormal urinary system	−0.05866095	84	6.468e−02	7.668e−01	FGF8:61 KIF26B:191 PPP3R1:296 BCL2:413 AGTR2:469 ETV4:535
MP0003948 abnormal gas homeostasis	−0.03335723	284	5.817e−02	7.668e−01	ATP2A1:15 ADRB1:57 FGF8:61 PASK:110 MB:126 GHRL:161
MP0003632 abnormal nervous system	0.05630742	93	6.228e−02	7.668e−01	CD86:18 DST:49 MSR1:214 PROM1:223 LRRK2:400 UHRF1:653
MP0005025 abnormal response to	0.03056415	346	5.690e−02	7.668e−01	PTPN13:53 IL6ST:71 STAT2:114 TNFSF10:174 CSF2RB:187 CXCR2:212

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
EBV.lymphocyte	0.057306574	669	1.026e−06	5.541e−05	CD86:18 SLFN14:20 ORC1:30 IRAG2:35 CD80:39 TMEM131L:55
nucleus.accumbens	−0.095087328	99	1.137e−03	3.069e−02	KCNA4:68 OTOF:69 HTR1D:75 NECAB2:241 PENK:270 PCDH11X:290
cerebellum	−0.042964669	374	5.089e−03	6.886e−02	TRANK1:1 SSTR4:13 SPHKAP:71 APC2:78 GLRA4:197 HID1:238
putamen	−0.208979784	15	5.101e−03	6.886e−02	RGS9:292 GRM3:781 GBX2:814 SLC18A3:999 DIPK1C:1033 HPCA:1152
spleen	0.042826522	298	1.219e−02	1.317e−01	SLFN14:20 NLRC5:44 VCAM1:125 BTN3A3:146 CSF1R:243 STAB1:373
Brodmann.area.9	−0.046912154	180	3.137e−02	1.856e−01	CHRNB2:19 HTR3B:41 PCDH8:137 RNF157:145 CALHM1:267 DLGAP2:277
cerebral.cortex	−0.061225771	107	2.947e−02	1.856e−01	CHRNB2:19 HTR3B:41 ADGRA1:106 KCNF1:115 CALHM1:267 C1QL2:284
esophagus.mucosa	−0.041833366	220	3.437e−02	1.856e−01	IQANK1:10 RNF222:29 CDKN2B:85 FAM83H:109 PADI3:113 MROH6:118
liver	0.035078862	327	3.187e−02	1.856e−01	AFP:14 MOGAT1:33 HSD17B2:76 PRODH2:112 AFM:123 KNG1:141
sigmoid.colon	−0.092513111	50	2.395e−02	1.856e−01	CHRNA3:48 TACR2:121 GJC1:243 PLA2G2C:735 SLC5A7:1047 CSRP1:1241
breast	−0.149891866	15	4.456e−02	2.188e−01	ALX4:210 THRSP:527 TNN:718 CIDEC:2523 LEP:2801 ABCB5:2830
blood	0.027813307	274	1.179e−01	4.547e−01	MEFV:118 CSF2RB:187 CXCR2:212 TLR1:327 OR2W3:432 TCL1A:434
subcutaneous.adipose	−0.053772296	72	1.158e−01	4.547e−01	INHBB:47 ACVR1C:149 APOLD1:255 ACACB:434 THRSP:527 LIPE:542
thyroid	0.038839204	150	1.031e−01	4.547e−01	CLCNKA:23 PLA2R1:64 LCN12:106 CLUL1:176 RAP1GAP:479 COL4A3:503
endocervix	−0.051925842	72	1.289e−01	4.641e−01	NXNL2:93 ZNF516:159 PWP2:174 KIF26A:182 BDKRB1:324 ELFN1:382
leg.skin	−0.028657332	217	1.499e−01	5.060e−01	RNF222:29 POF1B:38 C2CD4D:81 FAM83G:122 BNC1:130 MAB21L4:152
hypothalamus	−0.049221437	66	1.680e−01	5.336e−01	PCDH15:3 LY6H:53 SYNGR3:308 DLX2:328 GRM7:712 HAP1:764
amygdala	−0.116121759	9	2.278e−01	6.835e−01	PCDH15:3 BCAN:2208 SLC17A7:2321 NEUROD6:3506 LHFPL3:8951 NEUROD2:8951
pancreas	0.034212195	94	2.537e−01	6.851e−01	FBXW12:3 CATSPERB:10 G6PC2:87 PRODH2:112 CELA3A:131 F11:340
skeletal.muscle	−0.022035726	238	2.470e−01	6.851e−01	ATP2A1:15 MB:126 SMTNL2:176 RGS9BP:179 OBSCN:229 MYBPC2:236
testis	0.008740600	1375	3.134e−01	7.691e−01	ZSCAN5B:1 CATSPERE:6 RGPD8:12 ZSCAN5A:13 HEATR4:26 CATSPERG:32
vagina	−0.030985982	90	3.117e−01	7.691e−01	RNF222:29 BNC1:130 MAB21L4:152 SDR9C7:173 EVPL:268 SALL3:321
C1.spinal.cord	−0.019970063	182	3.570e−01	8.381e−01	ACSBG1:54 ENPP2:76 S1PR5:154 BACE1:181 PADI2:303 PLCL1:510
adrenal.gland	−0.023294375	108	4.053e−01	8.534e−01	MAP3K15:39 KLHL4:91 GIPC2:228 SLC47A1:343 CELA1:384 CYB561A3:749
cerebellar.hemisphere	−0.011665102	500	3.837e−01	8.534e−01	SSTR4:13 MINAR1:27 PER3:51 SPHKAP:71 KIF26B:191 GALNT9:219
ectocervix	0.046751412	25	4.189e−01	8.534e−01	PAPPA2:972 MMRN1:1367 EDNRA:1723 GDF7:7017 TRHDE:7017 ADAMTSL5:7017
ovary	−0.022885150	102	4.267e−01	8.534e−01	SUSD4:31 MMP21:98 ADAMTS1:112 GLI2:147 KLHDC8A:663 ADAMTS16:849
peyers.patch	−0.017103585	167	4.493e−01	8.665e−01	TRPM5:16 GPR35:99 EPS8L3:136 SLC17A4:144 TREH:203 CCR9:280
stomach	−0.022755511	86	4.675e−01	8.705e−01	MYH13:18 CAPN8:44 EPS8L3:136 GHRL:161 ATP4B:320 TM4SF5:441
substantia.nigra	−0.056064665	12	5.015e−01	9.027e−01	KCNE5:1480 SLC6A3:2569 FOXB1:2804 SPX:3163 CHRM5:8951 DBX2:8951
aorta	−0.008896225	154	7.053e−01	9.159e−01	COL8A2:90 PCDH11X:290 LOXL1:311 OGN:431 ST6GAL2:466 PTGIR:515
bladder	−0.023872570	24	6.859e−01	9.159e−01	PADI3:113 UPK2:428 RD3:1784 UGT1A6:1918 MSX2:2365 DHRS2:3651
esophagus.muscularis.mucosa	−0.023172547	26	6.829e−01	9.159e−01	SYNM:36 FOXF2:1055 FAM83D:1077 CHRM3:1081 NTN1:2441 EPHA7:2505
fallopian.tube	0.012103415	52	7.633e−01	9.159e−01	RARRES1:240 BCHE:905 GREB1L:1192 CDHR4:1251 GAS2L2:1258 HOXA3:1450
greater.omentum	−0.015614288	33	7.566e−01	9.159e−01	BNC1:130 THRSP:527 SOX17:1324 ALOX15:1546 CIDEC:2523 LEP:2801
hippocampus.proper	−0.031839323	10	7.275e−01	9.159e−01	OLIG2:2183 SLC17A7:2321 NEUROD6:3506 GRP:3678 NEUROD2:8951 HRK:8951
left.ventricle	−0.011050832	68	7.534e−01	9.159e−01	MB:126 ADPRHL1:433 SCN5A:536 MYH7B:774 POPDC2:1641 MYZAP:1855
pituitary.gland	−0.009361742	213	6.411e−01	9.159e−01	RIMBP2:9 GPR68:142 MYO15A:155 KCNH2:189 C2orf50:263 RNF212:265
suprapubic.skin	−0.006996168	193	7.398e−01	9.159e−01	RNF222:29 MFSD2A:50 BNC1:130 MAB21L4:152 SDR9C7:173 EVPL:268
tibial.artery	−0.012647359	106	6.544e−01	9.159e−01	GPR20:247 COL18A1:405 CSPG4:419 CDH6:746 FOXC2:1090 EFHD1:1219